

CSE 309: Midterm Sample Questions

Question 1 [35 points]

Give regular expressions for the following languages over the alphabet $\Sigma = \{\mathbf{a}, \mathbf{b}\}$.

- (a) (5 points) All valid integers.
- (b) (5 points) All valid real numbers.
- (c) (5 points) All valid complex numbers.
- (d) (5 points) All strings that odd number of **a**'s and even number of **b**'s.
- (e) (5 points) All strings that contain **aba** as a substring followed by **b**'s in multiple of 3.
- (f) (5 points) All strings that zero or more copies of **ab** or **ba**.
- (g) (5 points) All strings that have an **a** at the second position and a **b** at the third last position.

Question 2 [15 points]

Prove that the following languages are not regular.

- (a) (5 points) $\{\mathbf{a}^{n-2}\mathbf{b}^n\mathbf{c}^{n+2} : n \geq 2\}$ for $\Sigma = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$.
- (b) (5 points) $\{0^p : p \text{ is prime}\}$ for $\Sigma = \{0\}$.
- (c) (5 points) Language of matching parenthesis pairs for $\Sigma = \{ (,) \}$.

Question 3 [5 points]

Convert following context-free grammar to Chomsky Normal Form:

$$\begin{aligned} A &\rightarrow \mathbf{aBCDd} \mid CDE \mid \epsilon \\ B &\rightarrow BBC \mid \mathbf{abc} \mid \epsilon \\ C &\rightarrow \mathbf{bb} \mid \epsilon \\ D &\rightarrow \mathbf{cde} \mid \epsilon \\ E &\rightarrow \mathbf{e} \mid \epsilon \end{aligned}$$

Question 4 [5 points]

Design a context-free grammar for the language $\{\mathbf{a}^i\mathbf{b}^{n-i} : i \geq 0, n \geq 0\}$.

Question 5 [5 points]

Design a context-free grammar for the language that allows all valid arithmetic expressions with additions, subtraction, multiplication, and division of integers.